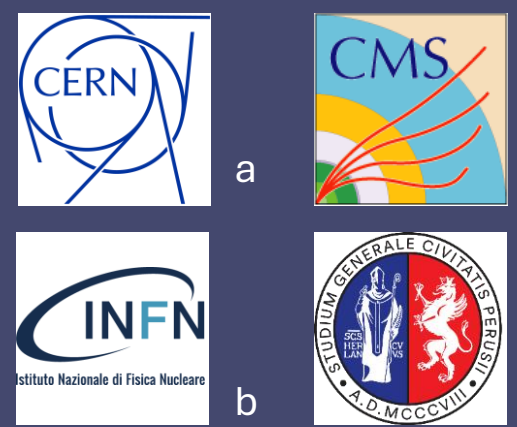




Thermal Quality Control of TB2S Ladders for the CMS Phase-2 Tracker Upgrade

Lorenzo Bistoni ^(a), Pier Filippo Cianchetta ^(a)

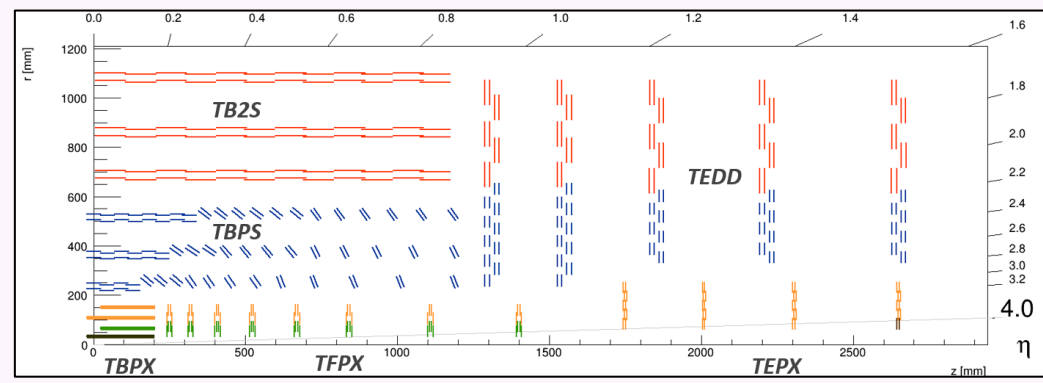
Contributions from: Duccio Abbaneo ^(a), Mohsin Abbas ^(a), Arsalan Ahmad ^(e), Ashfaq Ahmad ^(d), Jeremy Andrea ^(c), Muhammad Irfan Asghar ^(d), Giorgio Baldinelli ^(b), Mirza Azam Beg ^(e), Francesco Bianchi ^(b), Joan Buchmann ^(c), Eric Chabert ^(c), Laurent Olivier Gross ^(c), Clement Haas ^(c), Munir Ahmed Khan ^(a), Meena Meena ^(c), Saleh Muhammad ^(d), Antti Onnela ^(a), Muhammad Asad Rafique ^(e), Gourab Saha ^(b), Lea Stockmeier ^(f), Cristiano Turrioni ^(b), Giovanni Zevi Della Porta ^(a). On behalf of the CMS GROUP

Forum on Tracking Detector Mechanics 2026, Elba, Italy

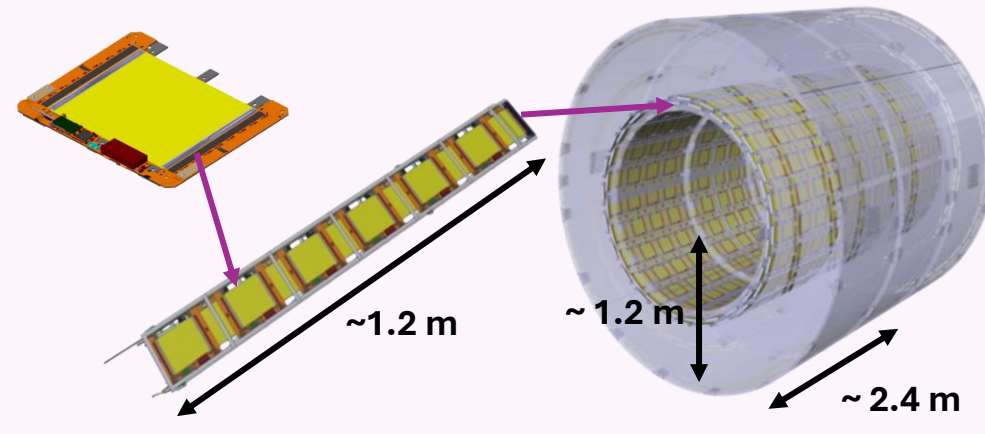


TB2S: Tracker Barrel with 2S modules

- 4416 2S type modules supported by 368 Ladders (12 each)
- Ladders manufactured and tested at New-2 and NCP in Pakistan
- 2S type modules integrated on Ladders at IPHC in Strasbourg
- Ladders integrated to wheel at CERN



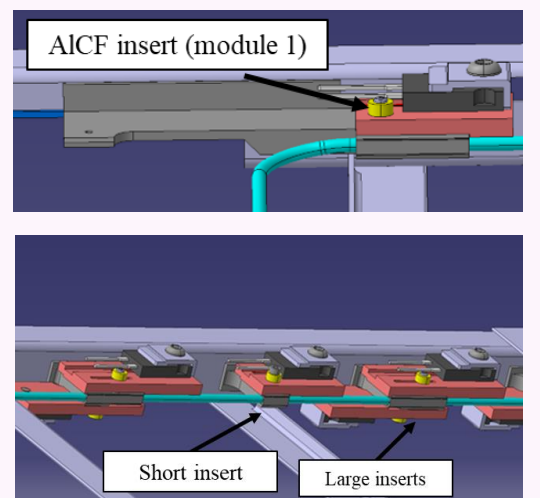
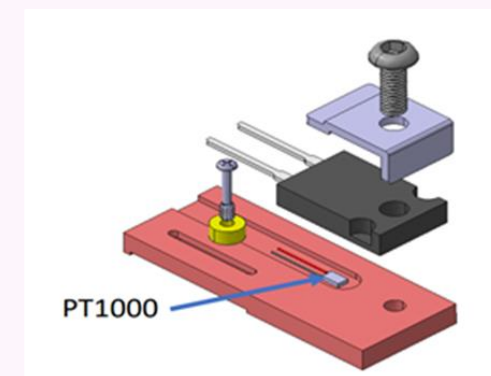
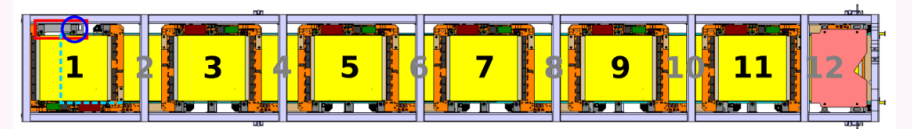
Layout of one quarter of the Tracker Upgrade for HL-LHC



From 2S module to TB2S wheel

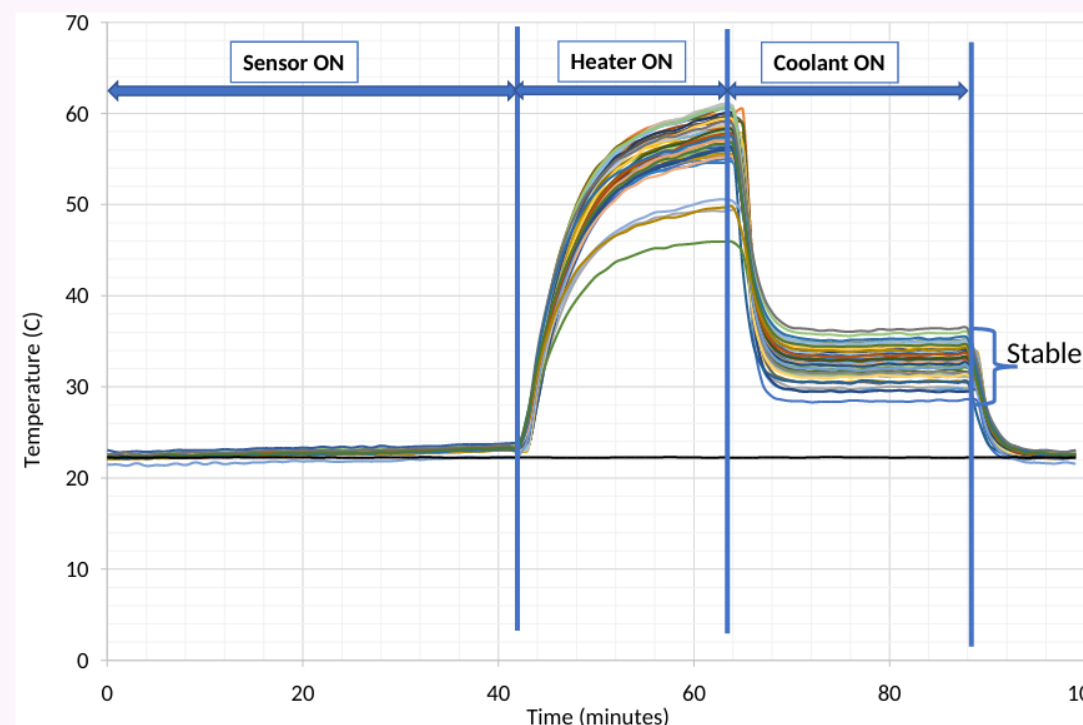
Thermal Quality Control strategy

- Module Cooling Points (inserts):**
 - 5 or 6 cooling points for modules 2 to 12
 - 4 cooling points for module 1
- Qualification of each insert:**
 - Heaters screwed in cooling points
 - 3 different insert types
- TQC Tests:**
 - Full production at New-2/NCP: water at +20°C
 - Pre-production Ladders (qty 4) at CERN: 2-phase CO₂ at +15°C and -35°C

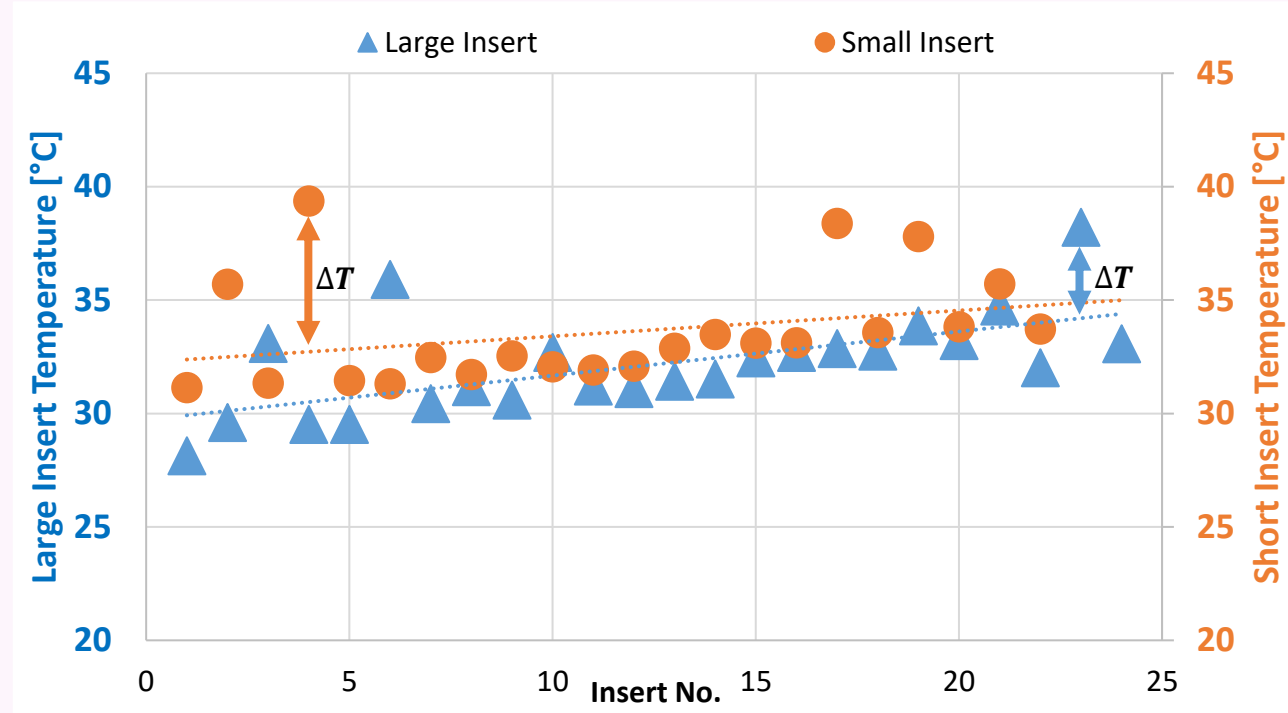


Testing procedure

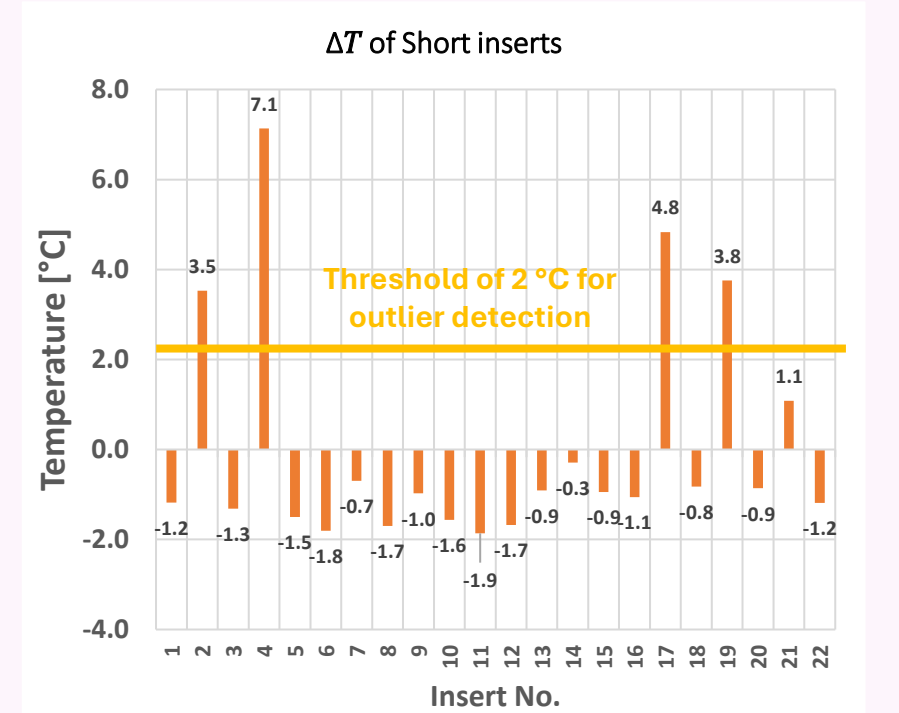
- Install heaters** (screw torque 6 cNm)
- Circulate coolant**
- Power heaters** (~1 W per heater)
- With stable temperatures, **compute ΔT** between insert and average value fitting line of this insert type
- Inspect** inserts with $\Delta T > 2^\circ\text{C}$ to check eventual problems



Heater temperatures during TQC cycle



Heater temperatures and fitting line extrapolation for one Ladder

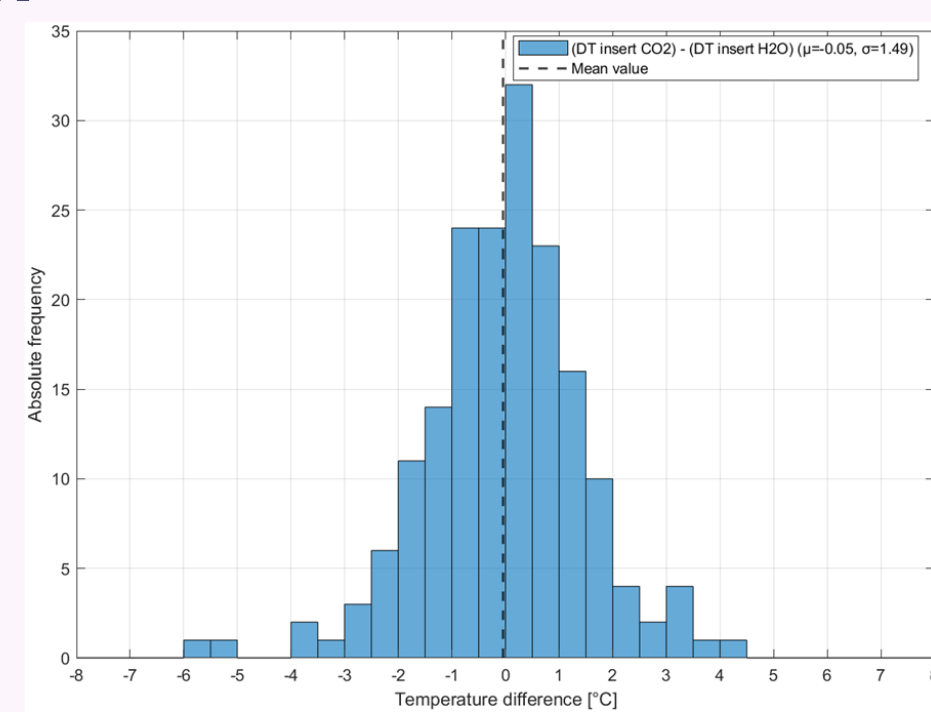


Compute ΔT for outlier detection

Water vs CO₂ comparison

- Same testing procedure**
- $\Delta T_{H2O} - \Delta T_{CO2}$:
 - Mean value** ~ 0: no bias between the H₂O and CO₂ TQC tests and comparable results
 - Std value** ($\sigma = 1.49^\circ\text{C}$): for a good ladder confirms that $\Delta T > 2^\circ\text{C}$ is a reasonable threshold for anomaly inspection

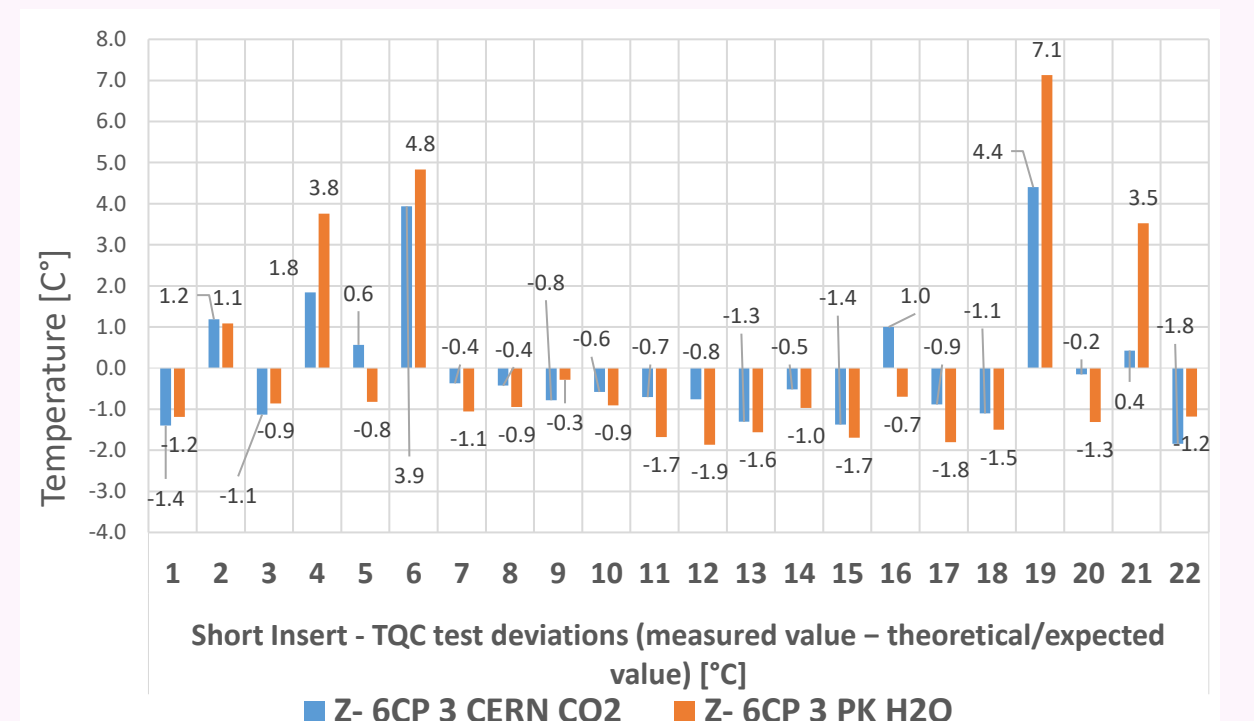
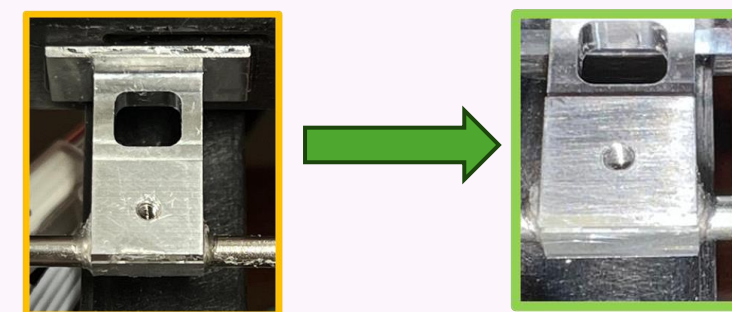
Mean [°C]	St_dev [°C]
0.05	1.49



Statistics collected for 4 Ladders

Anomalies spotted

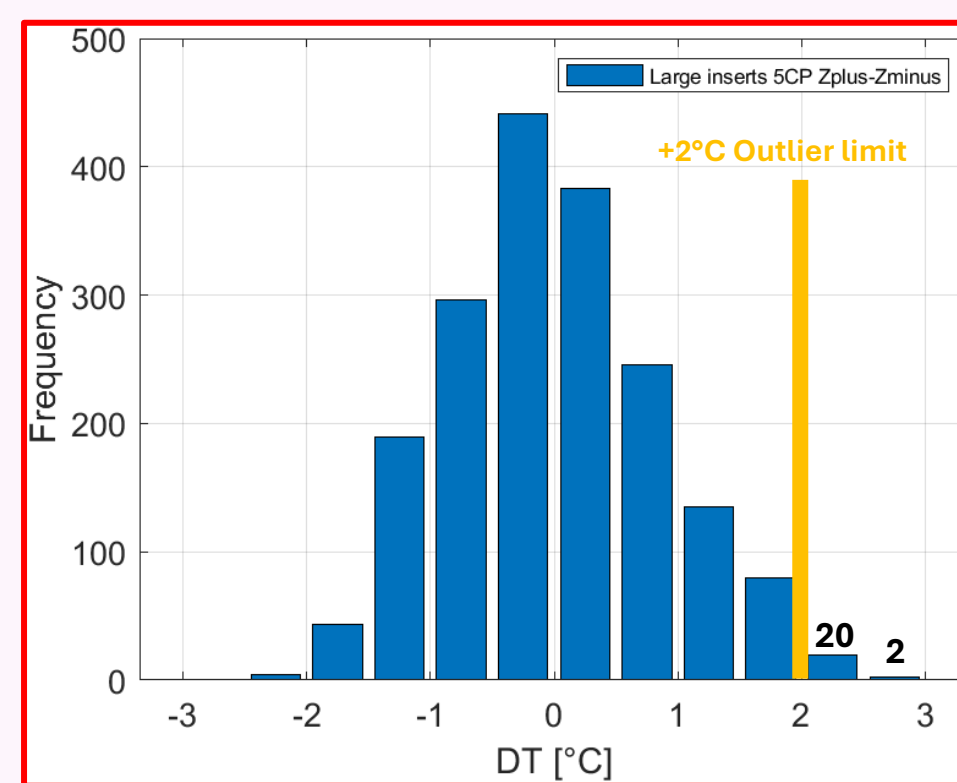
- Same anomalies have been detected in both CO₂ and H₂O tests (**excess of glue on short inserts of preprod. Ladders – fixed!**)
- The TQC procedure **detected outliers** and **triggered needed repairs**



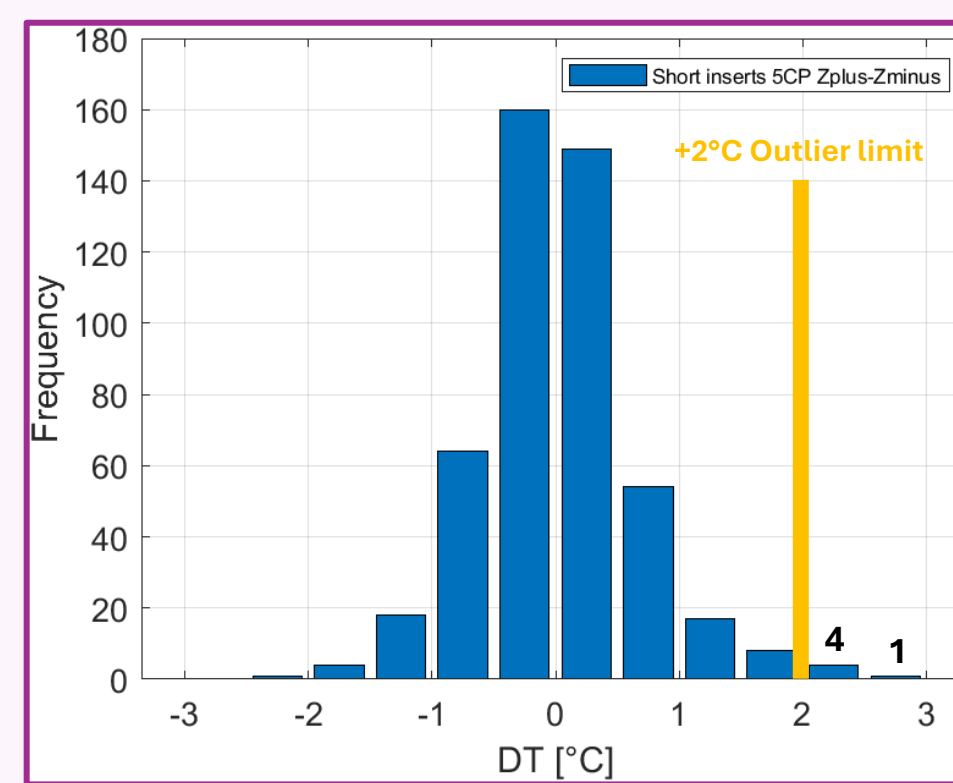
Comparison TQC tests CERN CO2 vs PK H2O / Z-6CP 3 small inserts

Production TQC

- Production Ladders tested: (40/368)**
 - 14 Ladders with outliers (9 with 1 anomaly, 1 with 2, 4 between 3 and 6)
- Outliers amongst large inserts: 22/1840 (1.2%)**
- Outliers amongst short inserts: 5/480 (1.0%)**
- Only few anomalies** $\Rightarrow$ **excellent thermal quality**



5CP short inserts



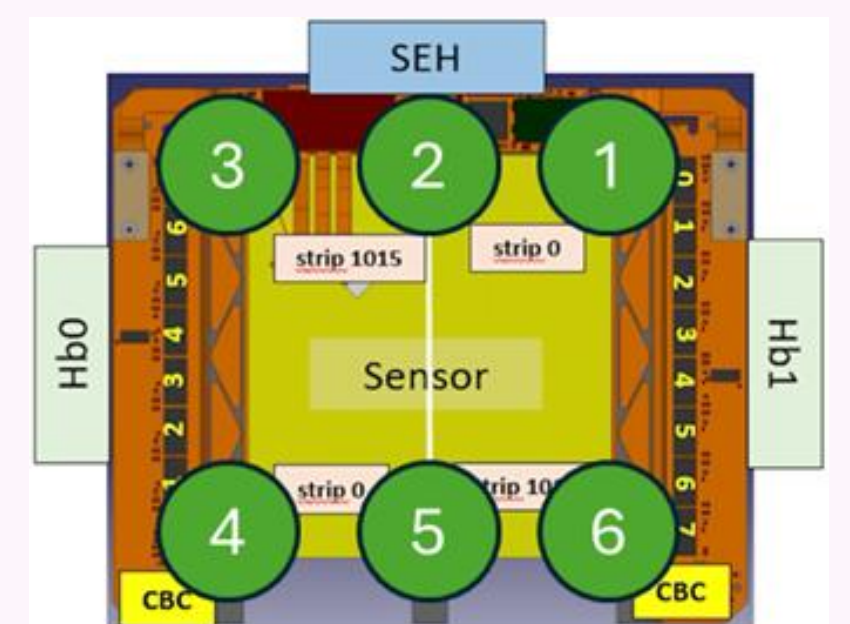
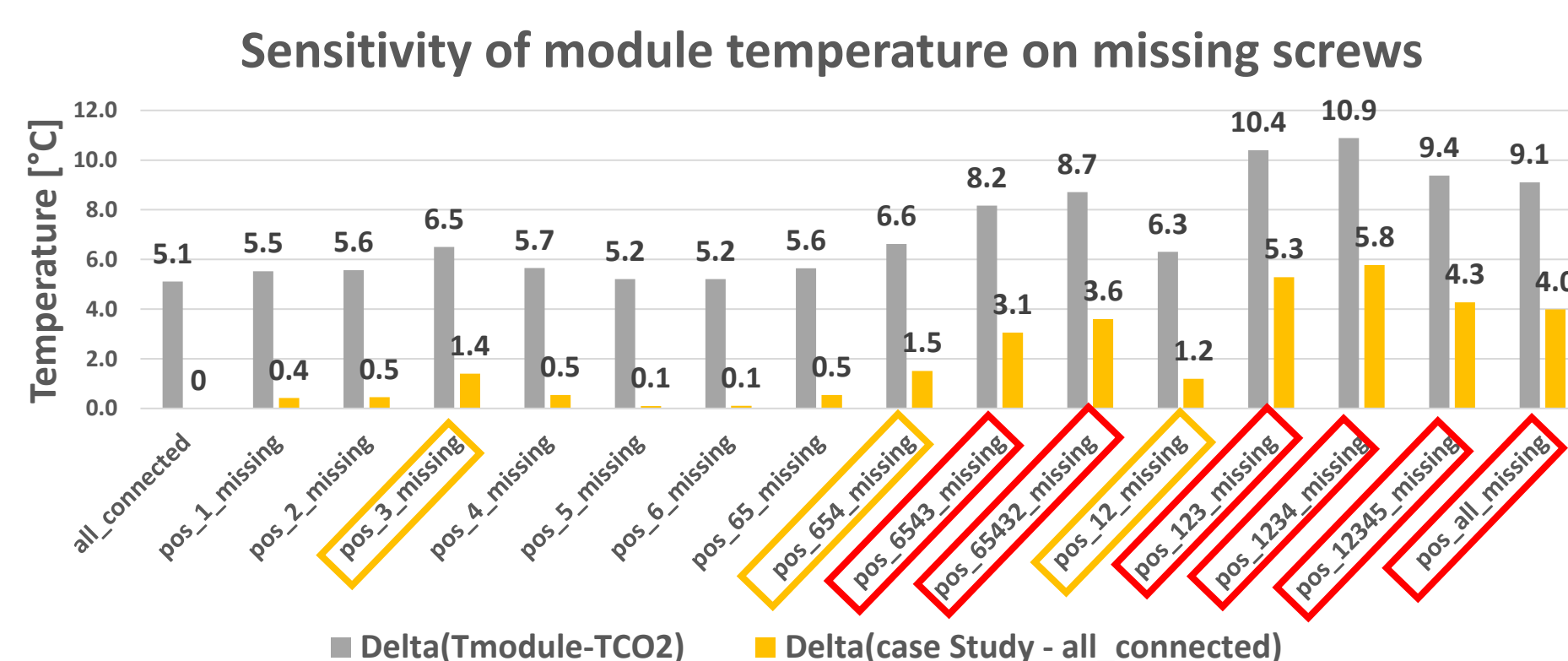
5CP short inserts



TQC setup at New-2 Ladder production facility

Thermal sensitivity measurements

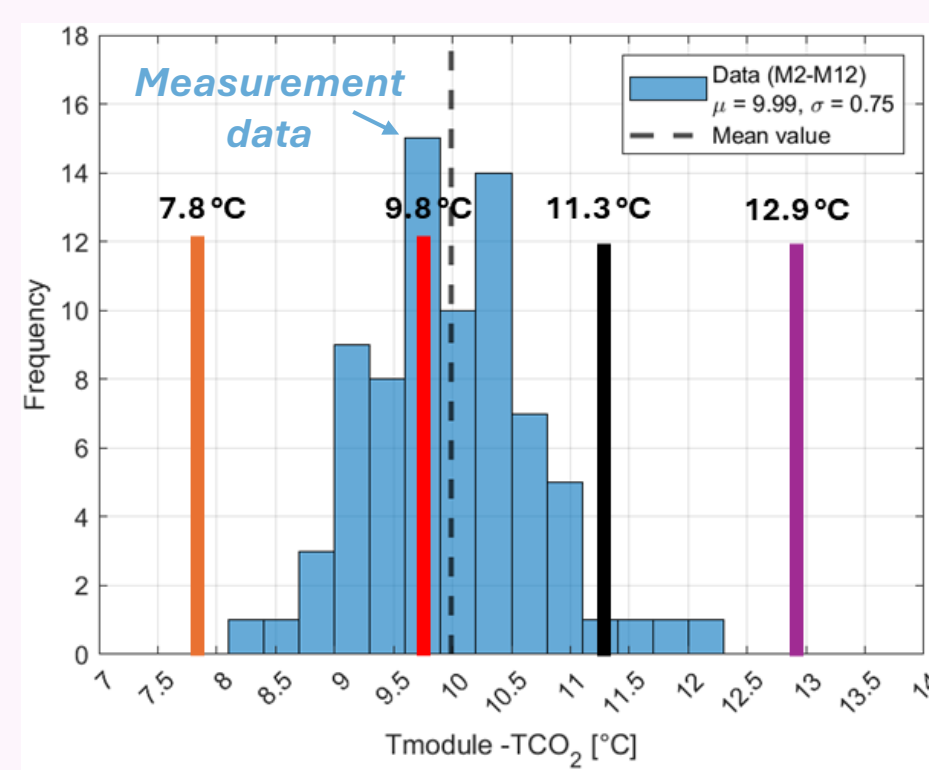
- Performed with real modules in Strasbourg with CO₂ at +15°C
- Reproducing bad thermal contact:** unscrewing one or multiple cooling contact screws of 6CP module
- Outliers don't necessarily compromise sufficient module cooling:** other inserts can compensate



Module attachment/cooling insert positions

Tests of fully assembled Ladder with modules

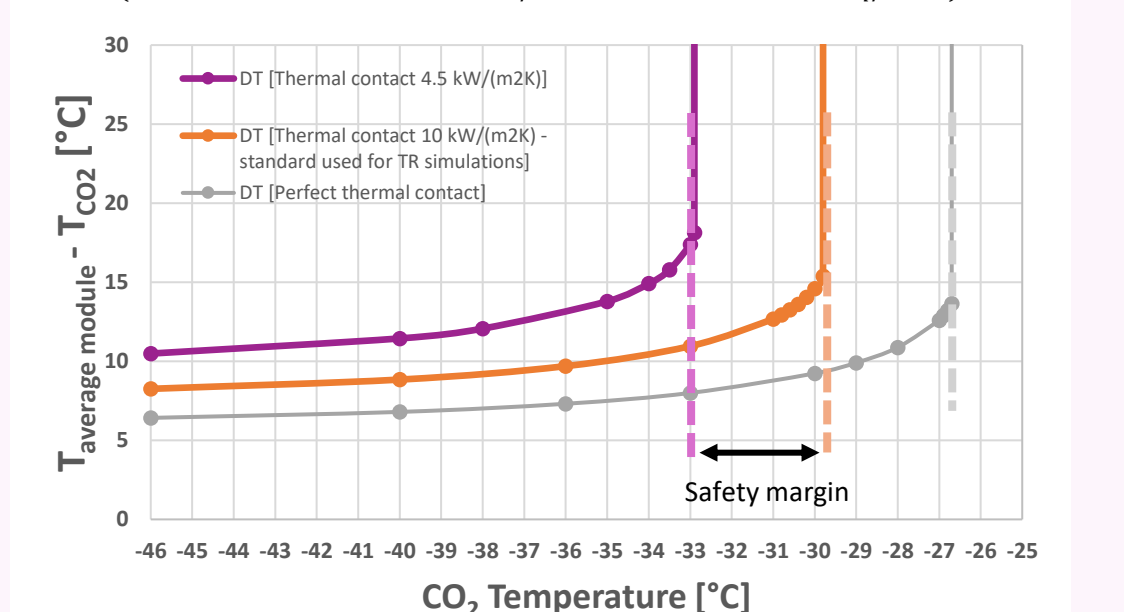
- Thermal test done in Strasbourg on all Ladders after final 2S module integration**
- Bottom sensor temperature recorded for each module
- Measure ΔT ($T_{\text{module}} - T_{\text{CO2}}$) with CO₂ at -30°C, T_{air} = -10°C and module power on
- Tests performed in test box containing air: **convective heat exchange** (in simulation use $HTC_{\text{air}} = 7 \frac{\text{W}}{\text{m}^2\text{K}}$)



Simulations:

- Adiabatic conditions and standard thermal contact (TC), 10 $\frac{\text{kW}}{\text{m}^2\text{K}}$ between module and inserts: $\Delta T = 7.8^\circ\text{C}$**
- Convection and perfect TC: $\Delta T = 9.8^\circ\text{C}$**
- Convection and standard TC: $\Delta T = 11.3^\circ\text{C}$**
- Convection and worst acceptable TC, 4.5 $\frac{\text{kW}}{\text{m}^2\text{K}}$: $\Delta T = 12.9^\circ\text{C}$**
 - Worst acceptable TC:** thermal runaway reached with CO₂ at max operating temperature (-33°C), ultimate scenario, adiabatic conditions

Thermal Runaway Simulations varying Thermal Contact, position #2 to #12, 6 CP, HTC CO₂ = 5 kW/m²K², Adiabatic conditions, Ultimate Scenario: (U_{bias}=800 V, Alfa=5.14·10⁻¹⁷ A/cm, Fluence = 3.68·10¹⁴ neq/cm²)



Measured module temperatures on Ladders do match with thermal runaway calculation results and meet the requirements of operation in CMS